Martingales — Exam (set 2)

Name: ______________________ **Date:** ______________

Duration: 3 hours (short set, $\approx 2\,$h of work). You may use any result from the course by citing it precisely. The questions can often be solved independently. Marks out of 100 (indicative). *Answer in the space provided; the rigour of your justifications matters.*

Problem I — Records of an i.i.d. sequence: a martingale and a weak law of large numbers **(24 pts)**

(indicative time: 28 min)

Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables with *continuous* law (i.e. $\mathbb{P}(X_i = X_j) = 0$ for $i \neq j$), defined on $(\Omega, \mathcal{F}, \mathbb{P})$. For $n \geq 1$ we say that a *record* occurs at time n if $X_n > X_k$ for all $1 \leq k < n$; we set

$$R_n = \mathbf{1}\{X_n > X_k \ \forall\, 1 \leq k < n\} \qquad (\text{with } R_1 = 1),$$

the indicator of a record, and we write

$$N_n = \sum_{k=1}^{n} R_k$$

for the number of records among the first n observations. We take for granted the following classical fact: *for every $n \geq 1$, by exchangeability (the maximum of $X_1, \ldots, X_n$ is, almost surely, uniformly distributed over the n positions), R_n is a Bernoulli random variable with parameter $1/n$, and the variables $R_1, R_2, \ldots$ are independent.*

We set $\mathcal{F}_n = \sigma(R_1, \ldots, R_n)$ and we define, for $n \geq 1$,

$$M_n \; = \; N_n - H_n, \qquad \text{where} \qquad H_n = \sum_{k=1}^{n} \frac{1}{k} \ \ (\text{harmonic number}).$$

(a) Show that $(M_n)_{n\geq 1}$ is an $(\mathcal{F}_n)$-martingale.

(b) Compute $\mathbb{E}[N_n \mid \mathcal{F}_m]$ for $1 \leq m \leq n$, in terms of N_m, H_n and H_m.

(c) Compute $\mathrm{Var}(N_n)$. Deduce that

$$\frac{N_n}{\log n} \xrightarrow[n\to\infty]{} 1 \qquad \text{in probability.}$$

Problem II — The Gambling-Team Method and the Expected Waiting Time for the Pattern HTH (26 pts)

(indicative time: 30 min)

We toss a fair coin at each time $n = 1, 2, \dots$: let $X_1, X_2, \dots$ be i.i.d. random variables with $\mathbb{P}(X_n = \mathtt{P}) = \mathbb{P}(X_n = \mathtt{F}) = \frac{1}{2}$ ($\mathtt{P}$ = Heads, $\mathtt{F}$ = Tails). We are interested in the first time the pattern $\mathtt{PFP}$ (Heads, Tails, Heads, in this order) appears:

$$T \;=\; \inf\{\, n \geq 3 \;:\; (X_{n-2}, X_{n-1}, X_n) = (\mathtt{P}, \mathtt{F}, \mathtt{P}) \,\}.$$

We use the *gambling-team method.* For each $k \geq 1$, a new player arrives just before toss k with a fortune of 1 euro and bets on the pattern in the following way. He stakes everything on "toss k equals $\mathtt{P}$"; the game is fair, so if he wins his fortune is multiplied by 2 (otherwise he is ruined and leaves the game). If he won, he stakes everything again on "toss $k+1$ equals $\mathtt{F}$", then, upon a further success, on "toss $k+2$ equals $\mathtt{P}$". A player who has completed the subsequence $\mathtt{PFP}$ stops (and keeps his fortune). Let G_n be the cumulative net gain of *all* the players who entered up to toss n (sum of the final fortunes minus the total of the initial stakes), with $G_0 = 0$. It is given that $\mathbb{E}[T] < \infty$.

(a) Show that $(G_n)_{n \geq 0}$ is a martingale with respect to the natural filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, and that T is a stopping time for $(\mathcal{F}_n)$.

(b) At time T (the pattern has just appeared), give explicitly the total final fortune of the players still present, and deduce the value of G_T as a function of T.

(c) Carefully justifying the application of the optional stopping theorem to G_n and the time T, compute $\mathbb{E}[T]$.

Problem III — Random series with random signs: an L^2-bounded martingale **(25 pts)**

(indicative time: 30 min)

Let $(\varepsilon_n)_{n\geq 1}$ be a sequence of independent and identically distributed random variables, defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with

$$\mathbb{P}(\varepsilon_n = +1) = \mathbb{P}(\varepsilon_n = -1) = \tfrac{1}{2}.$$

Let $(a_n)_{n\geq 1}$ be a sequence of real numbers satisfying

$$\sum_{n\geq 1} a_n^2 < \infty.$$

Set $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and, for $n \geq 1$, $\mathcal{F}_n = \sigma(\varepsilon_1, \ldots, \varepsilon_n)$. Define

$$X_0 = 0, \qquad X_n = \sum_{k=1}^{n} a_k\, \varepsilon_k \quad (n \geq 1).$$

(a) Show that $(X_n)_{n\geq 0}$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$, and that it is bounded in L^2, that is $\sup_{n\geq 0} \mathbb{E}[X_n^2] < \infty$. (Express $\sup_n \mathbb{E}[X_n^2]$ in terms of the a_k.)

(b) Deduce that there exists a random variable X_∞ such that $X_n \to X_\infty$ almost surely and in L^2. Compute $\mathbb{E}[X_\infty]$ and $\mathbb{E}[X_\infty^2]$.

(c) For $N \geq 1$, let $S_N^* = \max_{0\leq n\leq N} |X_n|$. Using Doob's L^2 maximal inequality, show that

$$\mathbb{E}\left[\left(\sup_{n\geq 0} |X_n|\right)^2\right] \leq 4 \sum_{k\geq 1} a_k^2.$$

In particular $\sup_{n \geq 0} |X_n| < \infty$ almost surely.

Problem IV — Exit Times and the Reflection Principle for Brownian Motion **(25 pts)**

(indicative time: 30 min)

Let $(B_t)_{t\in\mathbb{R}_+}$ be a standard Brownian motion in $\mathbb{R}$, started from $B_0 = 0$, and let $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$ be its filtration. Recall (without reproving them) that the processes $(B_t)_{t\in\mathbb{R}_+}$ and $(B_t^2 - t)_{t\in\mathbb{R}_+}$ are $(\mathcal{F}_t)$-martingales.

Fix a real number $a > 0$ and set

$$\tau \;=\; \inf\{t \in \mathbb{R}_+ \;:\; |B_t| \geq a\},$$

the first exit time from the open interval $]-a, a[$ (with the convention $\inf \varnothing = \infty$).

We will admit the following two facts:

(F1) $\tau < \infty$ almost surely, and $\mathbb{E}[\tau] < \infty$;

(F2) $B_\tau \in \{-a, a\}$ almost surely (continuity of the trajectories).

1. *(Symmetry argument, without optional stopping.)* Admit that the process $(-B_t)_{t\in\mathbb{R}_+}$ is also a standard Brownian motion with filtration $(\mathcal{F}_t)$. Noting that τ is unchanged when B is replaced by $-B$, show that

$$\mathbb{P}(B_\tau = a) = \mathbb{P}(B_\tau = -a) = \tfrac{1}{2}.$$

2. By applying the optional stopping theorem to the martingale $(B_t^2 - t)_{t\in\mathbb{R}_+}$ at the bounded stopping times $\tau \wedge t$, and then passing to the limit $t \to \infty$ (carefully specifying the convergence theorem used for each of the two sides), compute $\mathbb{E}[\tau]$ as a function of a.

3. Recall the *reflection principle* for Brownian motion: for every $t > 0$ and every $b > 0$,

$$\mathbb{P}\Big(\max_{s\in[0,t]} B_s \geq b\Big) \;=\; 2\,\mathbb{P}(B_t \geq b).$$

Fix $t > 0$. Using this principle and the scaling invariance of Brownian motion, show that

$$\mathbb{P}\Big(\max_{s\in[0,t]} B_s \geq b\Big) \;=\; 2\Big(1 - \Phi\big(b/\sqrt{t}\big)\Big) \qquad (b > 0),$$

where Φ denotes the cumulative distribution function of the $\mathcal{N}(0,1)$ law. Deduce, setting $M_t := \max_{s\in[0,t]} B_s$, that M_t has the same law as $|B_t|$, and give the density of M_t.